Matthias Grenié

Assistant Professor

Curriculum Vitae
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Université Grenoble-Alpes, Laboratoire d'Écologie Alpine (LECA, UMR 5553)
rekyt.github.io
+49 341 9733176
matthias.grenie@gmail.com
@LeNematode
Rekyt
fZ1_d7QAAAAJ
Matthias-Grenie
0000-0002-4659-7522

Academic Education

2016-2020 **PhD in Ecology** — École Doctorale 584 GAIA, Univ. Montpellier
Montpellier, France
- Supervised by **Pr. Françoiz Munoz** and **Dr. Cyrille Violle**
  Title: 'Outside the norm: deviation from ecological optimality and functional originality'

2013-2015 **MSc. in General Biology** — École Normale Supérieure de Lyon
Lyon, France

2012-2013 **Bsc. in General Biology** — École Normale Supérieure de Lyon
Lyon, France

2010-2012 **Prep. Classes for 'Grandes Écoles'** — Lycée Saint-Louis
Paris, France
- Biology, Geology, Mathematics, Physics, and Chemistry

Academic Experience

Sept. 2023-**present** **Assistant Professor** — Université Grenoble Alpes (UGA), Laboratoire d'Écologie Alpine (LECA)
Grenoble, France
- Appointed at the Pharmacy Faculty to teach Applied Statistics

Oct. 2020-Aug. 2023 **Postdoctoral Researcher** — German Center for Integrative Biodiversity Research (iDiv) / Leipzig University
Leipzig, Germany
- Supervised by **Dr. Marten Winter** on the functional biogeography of alien plants

2016-2020 **PhD Student** — Univ. Montpellier / CEFE
Montpellier, France
- Supervisors: **Pr. François Munoz** and**Dr. Cyrille Violle**. Independent PhD funding

Feb. 2015-June 2015 **Master Intern** — EcoFOG
Kourou, French Guiana, France
- Supervisor: **Dr. Bruno Hérault** on intraspecific variability in tropical trees growth in function of functional traits

Sept. 2014-Dec. 2014 **Master Intern** — ISEM, University of Montpellier
Montpellier, France
- Supervisor: **Dr. Ophélie Ronce** on matrix population models

Feb. 2014-July 2014 **Master Intern** — Indiana University
Bloomington, IN, USA
- Supervised by **Dr. Jean-François Goût** in Michael Lynch's lab on trait motifs in the *Paramecium* complex

June 2013-Aug. 2013 **Bachelor Intern** — Université Pierre et Marie Curie Paris 6
Paris, France
- Supervisors: **Dr. Éric Bapteste** and **Prof. Philippe Lopez** on sequence similarity networks of the metabolic pathways of *Chlamydomonas reinhardtii* in function of their origin

Funding

2023-**now** **Funded Project** Flexpool (internal iDiv) 10k€
- ➤ Co-author and Co-PI of the project led by **Dr. Qiang Yang** on the impact of naturalized plant species on plant-pollinator networks across elevational gradients

2022-**now** **Funded Project** Flexpool (internal iDiv) 10k€
- ➤ Co-author and Co-PI of the project led by **Dr. Bettina Ohse** on the link between functional traits and demographic rates of trees

Sept. 2016 **Doctoral Scholarship** École Normale Supérieure de Lyon
- ➤ Specific Doctoral Scholarship for Students of the ENS de Lyon

Scientific Contributions

15 publications (5 as first author). 457 citations (Google Scholar). h-index 10 (Google Scholar). 6 R packages

Selected Publications

1. **Grenié, M.**, Bruelheide, H., Dawson, W., Essl, F., Van Kleunen, M., Kühn, I., Kreft, H., Pyšek, P., Weigelt, P., & Winter, M. (2025). Gaps in Global Alien Plant Trait Data and How to Fill Them. *Global Ecology and Biogeography*, *34*(10), e70131. https://doi.org/10.1111/geb.70131

2. Schrodt, F., Beck, M., Estopinan, J., Bowler, D. E., Fontaine, C., Gaüzère, P., Goury, R., **Grenié, M.**, Martins, I. S., Morueta-Holme, N., Santini, L., … Thuiller, W. (2025). Advancing causal inference in ecology: Pathways for biodiversity change detection and attribution. *Methods in Ecology and Evolution*, *16*(10), 2276–2304. https://doi.org/10.1111/2041-210X.70131

3. Pili, A. N., Leroy, B., Measey, J. G., Farquhar, J. E., Toomes, A., Cassey, P., Chekunov, S., **Grenié, M.**, van Winkel, D., Maria, L., Diesmos, M. L. L., … Chapple, D. G. (2024). Forecasting potential invaders to prevent future biological invasions worldwide. *Global Change Biology*, *30*(7), e17399. https://doi.org/10.1111/gcb.17399

4. **Grenié, M.**, & Gruson, H. (2023). Fundiversity: A modular R package to compute functional diversity indices. *Ecography*, *2023*(3), e06585. https://doi.org/10.1111/ecog.06585

5. **Grenié, M.**, Berti, E., Carvajal-Quintero, J., Dädlow, G. M. L., Sagouis, A., & Winter, M. (2022). Harmonizing taxon names in biodiversity data: A review of tools, databases and best practices. *Methods in Ecology and Evolution*, 2041–210X.13802. https://doi.org/10.1111/2041-210X.13802

R packages

- ➤ **`funrar`** (creator and maintainer)
  Compute functional rarity indices, major tool for the work of the FREE CESAB group
  published paper – CRAN – GitHub
- ➤ **`ecolottery`** (co-creatore)
  efficient simulation of community assembly processes, including both neutral and niche processes
  published paper – CRAN – GitHub
- ➤ **`fundiversity`** (co-creator and maintainer)
  compute functional diversity indices in an efficient way, modular, with modern computation features out-of-the-box (parallelization, memoization)
  published paper – CRAN – GitHub
- ➤ **`funbiogeo`** (co-creator and maintainer)
  ease functional biogeography analyses, developed within FREE CESAB group
  GitHub
- ➤ **`rtaxref`** (creator and maintainer)
  access the data of the French Taxonomic Reference (TAXREF) API
  GitHub
- ➤ Significant contributions to other R packages: **`taxize`** (extract and standardize taxonomic data), **`ggfortify`** (simplified and automated plotting), et **`traitdataform`** (metadata writing help for functional traits)